Sri Manaswi Tipparaju

Site Reliability Engineer • Cloud Platform Engineering • Infrastructure Automation

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SUMMARY

13+ years embedded inside two of the largest enterprise engineering organizations in the industry — Oracle and Cisco — keeping mission-critical platforms alive, fast, and scalable. Background sits at the intersection of performance engineering, cloud infrastructure, and automation: diagnosing production incidents, building the tooling that prevents recurrence, and running reliability programs across 20+ engineering teams at scale.

At Oracle, responsible for SLO/SLA targets on OCI-hosted cloud management platforms — building automated provisioning pipelines, observability setups, and the blameless post-mortem culture that actually reduces repeat incidents. Equally comfortable in the terminal writing Python and shell automation, triaging a P1 in a war room, or designing containerized architectures on multi-tenant OCI workloads.

Recent work extends into AI-powered tooling — PerfAgent is an autonomous diagnostic pipeline that classifies performance artifacts, runs parallel LLM-based analysis across 9 diagnostic domains, and routes severity-ranked findings to the right team in under 60 seconds.

AWS and Linux foundation certified with strong foundation in OCI, AWS, Oracle Autonomous DB, containerization, CI/CD and performance diagnostics (AWR, ADDM, AppDynamics, CloudWatch).

TECHNICAL SKILLS

SRE / Reliability	SLOs · SLAs · Error Budgets · Incident Management · On-Call Runbooks · Chaos Engineering · Capacity Planning · Toil Reduction
Cloud	OCI · AWS (EC2, EKS, RDS, S3, VPC, IAM, Lambda, CloudWatch) · Multi-cloud Architecture
Containers / IaC	Kubernetes (CKA) · Docker · Helm · Terraform · CloudFormation · CI/CD (CodePipeline, CodeBuild, Git)
Observability	CloudWatch · AppDynamics · OEM · AWR/ADDM/SQL Monitor · Alerting & Dashboarding · Custom Prometheus metrics · LLMOps instrumentation
Automation / Dev	Python · Shell · SQL/PL/SQL · TypeScript · Node.js · React · REST APIs · AI/LLM Integration
Databases	Oracle Autonomous DB · Oracle RDBMS (CDB/PDB Multitenant) · PostgreSQL · AWS RDS · Aurora
AI/LLM	Anthropic Claude API · OpenAI embeddings · Pinecone · Multi-agent orchestration · LLMOps · Token / cost optimization

EXPERIENCE

Site Reliability / Cloud Platform Engineer | Oracle Inc — Oracle Management Cloud & Enterprise Manager | *Mar 2017 – April 2026*

- Worked on SLO/SLA reliability for OCI-hosted Oracle Management Cloud across multi-region deployments — helped build out automated health checks and failover runbooks that brought platform availability to 99.9%+.
- P1 incidents were routinely taking 3+ hours to resolve with engineers jumping between multiple OEM dashboards — contributed to rebuilding the alerting pipeline with structured triage playbooks, which cut MTTR by ~40% within the first quarter.
- Database provisioning for new OCI environments was a manual 4–6 hour process — automated the workflow using Python and shell scripting, bringing it down to under 30 minutes with consistent environment parity across dev, staging, and production.
- Ran SQL/PL/SQL performance reviews across 20+ Oracle engineering teams where recurring p99 latency spikes were impacting shared OCI platform services — used AWR, ADDM, and SQL Monitor to diagnose bottlenecks and helped teams improve query throughput by 60%+ on the highest-traffic workloads.
- WebLogic thread pool saturation was causing SLA breaches on Enterprise Manager services — tuned thread pool and connection pool settings, improving response times by ~30%.
- A Fortune 10 customer needed Oracle Enterprise Manager migrated from Solaris to Linux with no tolerance for downtime — designed the cross-platform migration strategy using Data Pump techniques, coordinated the execution with the broader team, and delivered with zero data loss and no production impact.
- Worked on high availability architecture for Oracle environments using Data Guard and GoldenGate — helped configure standby databases and replication setups that kept critical OCI workloads resilient to failures and supported near-zero RPO/RTO targets.
- Recurring incidents had no structured follow-through and the same failure patterns kept resurfacing — helped introduce blameless post-mortems and PIR processes, which reduced repeat incidents and created a shared knowledge base used across teams.
- Contributed to integrating OpenAI Codex into the development workflow to auto-generate test cases from code and specs, meaningfully expanding coverage without proportional increase in manual effort.
- Helped incorporate Codex-assisted code review and test case suggestions into the CI cycle, catching issues earlier in the Jenkins pipeline and reducing the back-and-forth between dev and QA teams during release prep.

Database Performance & Reliability Engineer | Cisco Systems Inc | *Sept 2012 – Mar 2017*

- Kept 200+ production databases healthy for mission-critical Cisco applications — proactive monitoring, automated alerting, and fast triage when things went sideways.
- Built a performance snapshot framework that proactively detected inefficient SQL across hundreds of DB instances before they caused incidents, shifting the team from reactive firefighting to proactive tuning.

- Automated load testing with LoadRunner before every major release to catch regressions before they hit production; diagnosed bottlenecks using AWR, ADDM, SQL Trace, and TKPROF.
- Used AppDynamics for heap analysis and memory diagnostics to track down cascading performance issues impacting end users.

PROJECTS

OpsMind (formerly PerfAgent) — AI SRE Copilot Platform *Live Demo · GitHub*

- Built an autonomous AI SRE copilot (Node.js + TypeScript + Anthropic claude-sonnet-4.5) that ingests production diagnostic artifacts — AWR reports, SQL Monitor output, thread/heap dumps, JFR recordings, JMeter results, browser logs, stack traces, and **Kubernetes diagnostics** — classifies them, runs parallel domain-specific LLM analysis across **10 specialized skills**, correlates findings across artifacts into a unified causal chain, and dispatches severity-ranked output to Slack, Jira, and email in under 40 seconds.
- **Multi-agent architecture (5 agents):** Classifier (skill routing from content preview), Skill Analyzer (parallel Claude calls with 4–8 concurrency), Cross-Artifact Correlator (builds incident timeline + causal chain linking findings across files), Chat Agent (conversational follow-up on incidents), Reporter (severity-ranked notifications). Orchestrator coordinates the 6-step pipeline with Critical-severity override to #incidents channel.
- **RAG over operational runbooks:** Pinecone vector DB with OpenAI text-embedding-3-small (1536d) retrieves relevant runbooks and postmortems before every skill analysis. In-memory cosine-similarity fallback enables offline demo with zero external dependencies. Every analysis cites source runbooks in the UI.
- **LLMOps observability:** Prometheus instrumentation tracks every Claude call — input/output tokens per skill, P50/P90/P99 latency histograms, cost in USD ($3/M input, $15/M output), parse success rate, RAG retrieval latency. Pre-built Grafana dashboard ships in the docker-compose stack.
- **10 diagnostic skills** as structured system prompts with strict JSON output contracts — each finding requires an `evidence` field grounded in specific numbers from the input, making hallucinations immediately identifiable. Added Kubernetes skill for OOMKilled, CrashLoopBackOff, HPA failures, and resource limit analysis.
- **Production scale:** Stateless Express server on Docker / AWS ECS Fargate, robust JSON parser with 4-layer fallback (parse → repair → sanitize → regex extract), streaming responses for skill analysis, REST endpoints for `/analyze`, `/chat`, `/rag/ingest`, `/metrics`, `/webhooks/{pagerduty,opsgenie,grafana}`, `/ci/sql-gate`. Config-driven team routing — adding a new Slack channel or Jira project is one config edit.
- **Performance:** End-to-end triage of 5 diagnostic files in 25–40 seconds at ~$0.10 per incident (vs. 2–4 hours and ~$500 in senior engineer time for manual triage — a 200× time and 5,000× cost reduction).
- **Tech:** Node.js · TypeScript · Anthropic Claude · Pinecone · OpenAI Embeddings · Express · Prometheus · Grafana · Docker · AWS ECS · Next.js (portfolio integration)

AI-Powered Cloud Insights Dashboard Live Demo

- Built a full-stack observability dashboard with React, Next.js, and Groq LLM APIs that surfaces real-time operational insights — summaries, key signals, and recommended actions — alongside raw data.
- Designed a modular API layer with structured response rendering so the output (summary, key points, actions) is machine-readable and easy to pipe into alerting or runbook workflows.

CERTIFICATIONS & EDUCATION

Certifications: AWS Solutions Architect – Associate (exp. Apr 2029) • Certified Kubernetes Administrator (CKA) (exp. Apr 2028)
Education: M.S. Engineering, Lamar University (2012) • B.E. Engineering, Acharya Nagarjuna University